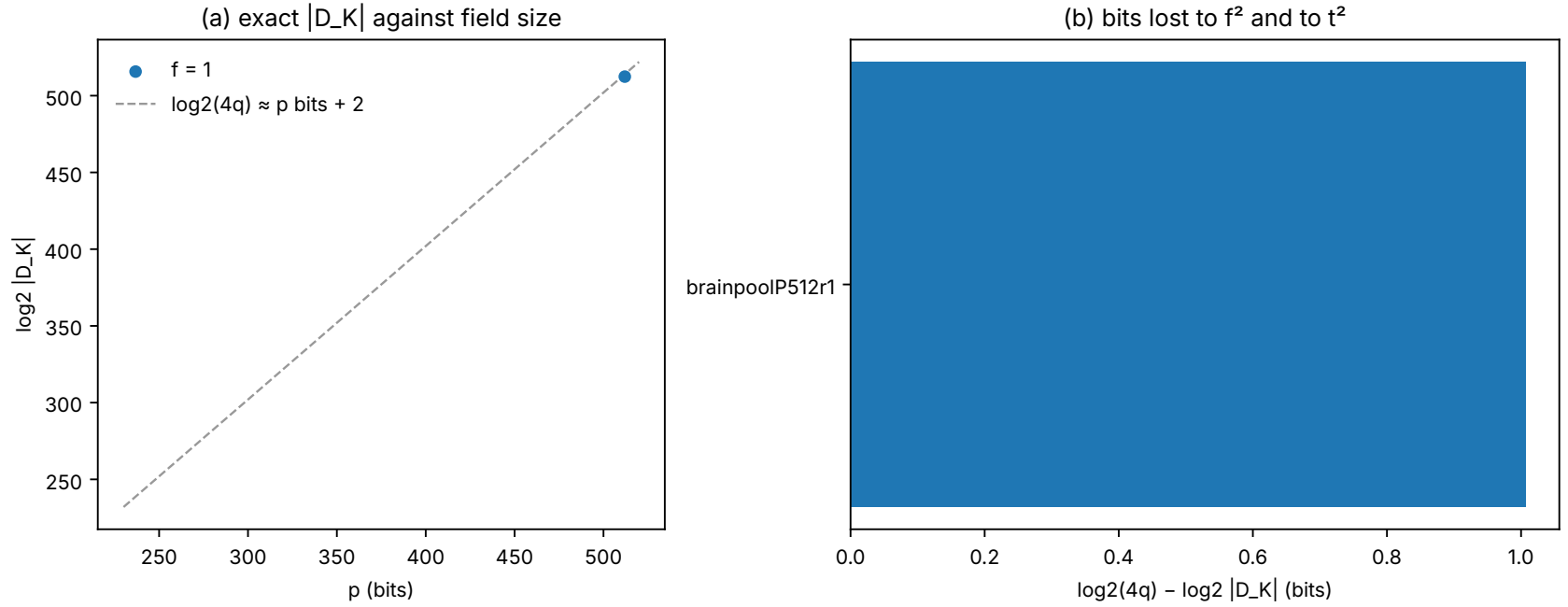
Exact CM discriminant of brainpoolP512r1 (batch 7), and the two composites still open

research/endosweep_dk_exact_batch7_20261106 (2026-11-06). Data: `dk_exact.json`, `factorizations.jsonl`, `open_composites.json`; std-curves commit `77fe6e3585ca2c2225b59d7df24b7c775437276f`. Every factor of $4q - t^2$ is proved prime by PARI `isprime`. These are exact computations, not measurements; the README states method, scope and regeneration.

Part A. brainpool/brainpoolP512r1 is the one curve completed between the sixth frozen driver snapshot (batch 6) and the seventh (2026-11-06). Its $4q - t^2$ (155 digits) is $557 \cdot p_{54} \cdot p_{98}$. The survey found 557; this project's driver split the remaining C152 into $p_{54} \cdot p_{98}$ with GMP-ECM 7.0.5 at B1 = 11e6 (2026-11-06T14:01:46Z, between 1650 and 1700 curves at that B1, far below the expected effort for a 54-digit factor). So $f = 1$ and $D_K = -(4q - t^2)$ with $4q - t^2 \equiv 3 \bmod 4$ (155 digits). std-curves records no `cm_disc`, so D_K was not checked against the literature. The factors were submitted to FactorDB after the split, which now lists the C152 and the full $4q - t^2$ as fully factored.



n = 1 curve; exact integers (PARI isprime-proved factorisations), no uncertainty.

Figure 1: (a) $\log_2|D_K|$ against p for brainpoolP512r1 ($f = 1$). (b) The bits between $\log_2 4q$ and $\log_2|D_K|$. Recorded values from `dk_exact.json`.

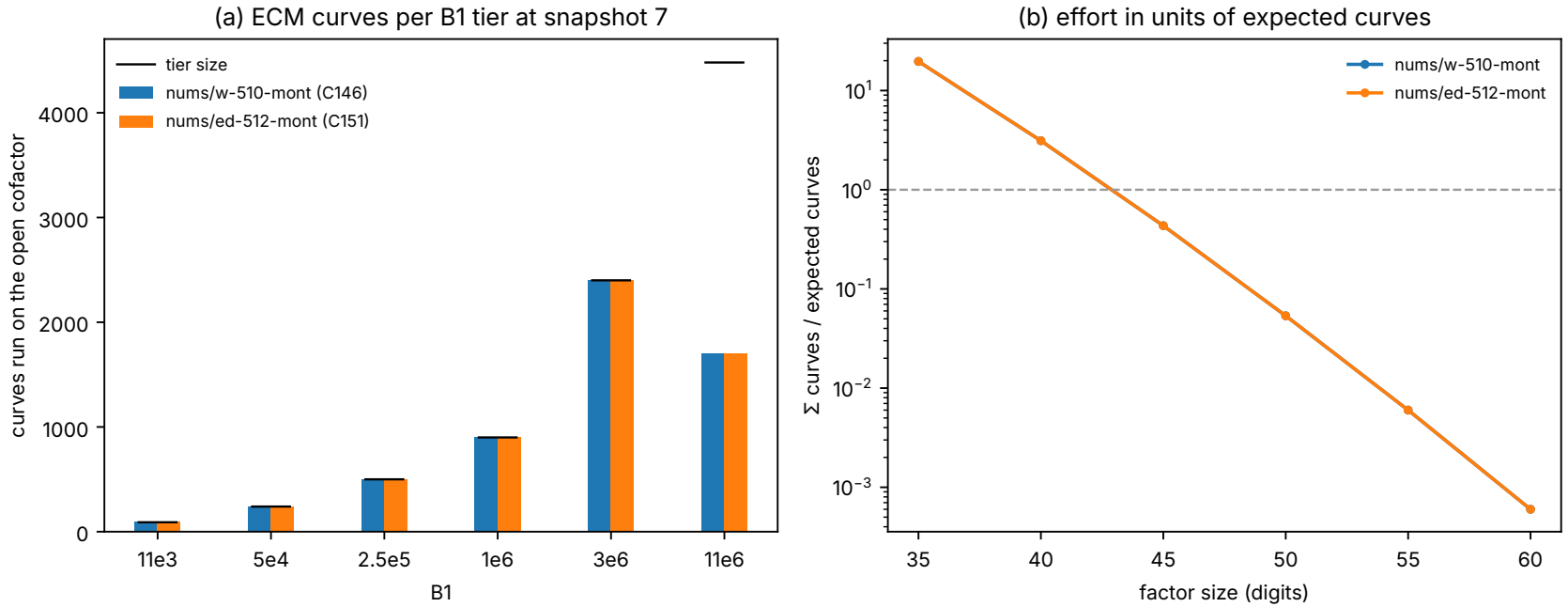
curve	p bits	4q−t² digits	s digits	f	D_K	log2 |D_K|	min degree	std-curves
brainpool/brainpoolP512r1	512	155	155	1	(155 digits)	512.41	2^510.41	no cm_disc

Lean: `CryptoResearch.Endosweep.DkExactBatch7.exact_discriminants` (target `TASK-20261106-966c20`), reusing `DkExact.DkCert`; fundamentality is conditional on the primality of the factors above 10^8 (PARI, not Lean).

Part B: the two $4q - t^2$ still open at snapshot 7

No D_K is stated for these. Curves per B1 are those the driver state records; the coverage columns divide them by GMP-ECM 7.0.5′s own expected-curve table (`ecm -v`) and give $\exp(-\sum)$, the approximate chance that a factor of that size, if one exists, was missed (an idealised model). The B1 = 11e6 counts rose from 1500 (snapshot 6) to 1700. The C146 and C151 are beyond practical NFS on this 4-core machine.

curve	cofactor	11e3	5e4	2.5e5	1e6	3e6	11e6 (of 4480)	Σ/E 40 dig.	Σ/E 45 dig.	FactorDB
nums/w-510-mont	C146	90	240	500	900	2400	1700	3.12	0.43	C
nums/ed-512-mont	C151	90	240	500	900	2400	1700	3.12	0.43	C



Counts: driver_state.snapshot7.json. Expected curves: GMP-ECM 7.0.5 ecm -v. Dashed: 1 expected find.

Figure 2: (a) ECM curves per B1 tier on each open cofactor at snapshot 7. (b) The same effort in units of GMP-ECM's expected number of curves per factor size; 1 (dashed) is one expected find.